

ζ -function method and the evaluation of fermion currents

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Using the ζ -function method, a prescription for the evaluation of fermion currents in the presence of background fields is given. The method preserves gauge invariance at each step of the computation and yields to a finite answer showing the relevant physical properties for arbitrary background configurations. Examples for $n = 2, 3$ dimensions are worked out, emphasizing the connection between preservation of gauge invariance and violation of other symmetries (chiral symmetry for $n = 2$, parity for $n = 3$).

I. INTRODUCTION

Evaluation of composite current operators (CCO) is of great interest for the analysis of different field theoretical models. As it is well known, the path-integral formulation of quantum field theory provides a natural framework for these calculations, in particular in connection with its topological aspects. Not only can the role of solitons be more easily understood through semiclassical approximations of the path integral, but, in addition, index theorems and other powerful mathematical tools can be put to work very simply.

When gauge theories are concerned, CCO computations are not only faced with divergencies—and hence with the necessity of regularization—but also with the need of preserving gauge invariance in the sense of Schwinger, who developed, in his 1951 classic paper,¹ a regularization procedure, the so-called fictitious-time method.

Schematically, this method consists of a point-splitting regularization of the originally divergent vacuum currents; since this spoils gauge invariance, a compensating phase factor is introduced *ad hoc*, in order to maintain this invariance at every stage of the computation. In some cases, like the exactly soluble QED₂ model² or when only constant or static fields in three dimensions are considered,^{3–5} the method is very simple. On the other hand, when non-Abelian gauge theories are investigated and arbitrary field configurations considered, it may become more involved.

It is the purpose of this work to propose an alternative method based on techniques developed by Seeley⁶ from his definition of complex powers of pseudodifferential operators. Not only is the application of this method very simple, but it is also automatically gauge invariant, thus avoiding, in particular, the introduction of phase factors. Moreover, isolation of relevant contributions to ground-state currents in general cases (i.e., nonconstant, nonstatic field strengths) is possible, as it will become clear from the examples we discuss.

II. PROCEDURES

Consider the generating functional (in Euclidean n -dimensional space) for fermions in the presence of a background field

$$Z[A] = \int \mathcal{D}\bar{\psi} \mathcal{D}\psi \exp\left(-\int \bar{\psi} \mathcal{D}(A) \psi d^n x\right) \\ \equiv \det \mathcal{D}(A), \quad (1)$$

with

$$\mathcal{D}(A) = \gamma_\mu (i \partial_\mu + A_\mu). \quad (2)$$

Here, $\mathcal{D}\bar{\psi} \mathcal{D}\psi$ is some measure in the space of fermion field configuration, to be defined below. The background field A_μ takes values in the Lie algebra of some gauge group G .

Of course, since the product of eigenvalues of the Dirac operator grows with no bound, Z needs a regularization in order to make sense. Disregarding this fact for a moment, one can try to define the ground-state current in the presence of the background field

$$J_\mu(x) = \langle \bar{\psi}(x) \gamma_\mu \psi(x) \rangle = -\frac{\delta}{\delta A_\mu(x)} \log Z[A] \\ = -\text{tr } \gamma_\mu G(x, x), \quad (3)$$

where the Green's function $G(x, y)$ satisfies

$$\mathcal{D}_x G(x, y) = \delta^{(n)}(x - y). \quad (4)$$

Expression (3) is also ill defined and it has to be regularized if we want to get a sensible result for J_μ . The method proposed by Schwinger^{1,2} makes use of a point splitting in order to deal with the problem of multiplication of distributions at the same point

$$J_\mu(x)_{\text{reg}} = \lim_{y \rightarrow x} \text{tr } \gamma_\mu G(x, y). \quad (5)$$

Here the limit has to be taken so that in Minkowski space $y \rightarrow x$ from the future and from the past. As noted by Schwinger, the appropriate gauge-invariant expression for the current is not (5), but

$$J_\mu(x)_{\text{pt}} = \lim_{y \rightarrow x} -\text{tr } \gamma_\mu G(x, y) \exp\left(-i \int_x^y A_\mu(z) dz\right). \quad (6)$$

The phase factor in Eq. (6) has been included in order to restore gauge invariance.

Let us now describe our alternative regularization method. As we stated above, the generating functional defined by Eq. (1) (the determinant of the Dirac operator $\mathcal{D}$) is an ill-defined object (the eigenvalues of $\mathcal{D}$ grow with no bound). This problem can be handled once and for all using the explicitly gauge-invariant ζ -function method⁷ (valid for any elliptic operator $\mathcal{D}$, not necessary Hermitian, defined on a compact manifold with no boundary⁸) in order to make sense from Z

$$Z_{\text{reg}}[A] = \exp\left(-\frac{d\zeta(\mathcal{D}(A), s)}{ds}\right)\Big|_{s=0}. \quad (7)$$

As expression (7) is well defined, the quantity

$$J_\mu(x)_{\text{reg}} \equiv -(\delta/\delta A_\mu(x)) \log Z_{\text{reg}}[A], \quad (8)$$

if the Z_{reg} proves to be differentiable, can be taken as the definition of the regularized ground-state current. As we shall see below, (7) is indeed differentiable and then one can easily obtain, from Eqs. (7) and (8), a regularized version of the last equality in (3). Indeed, let us write for the functional derivative

$$\frac{\delta Z_{\text{reg}}}{\delta A_\mu(x)}[A](a) = \frac{d}{dt}(Z_{\text{reg}}[A + ta_\mu] - Z_{\text{reg}}[A]), \quad (9)$$

where a_μ indicates the direction (in the space of A_μ) in which the functional derivative is taken. Now, from expression (7) and the results about differentiability of the ξ function,⁹ we can write

$$\begin{aligned} \frac{\delta}{\delta A_\mu} \log Z_{\text{reg}}[A](a) \\ = \frac{d}{dt} \left(-\frac{d\xi}{ds} (\mathcal{D}(A + ta_\mu), s) \Big|_{s=0} \right)_{t=0} \end{aligned} \quad (10)$$

or

$$\frac{\delta}{\delta A_\mu} \log Z_{\text{reg}}[A](a) = \frac{d}{ds} \{ s \operatorname{tr} (\mathcal{D}(A)^{-s-1} \gamma_\mu a_\mu) \} \Big|_{s=0}, \quad (11)$$

and hence the current $J_{\mu\text{reg}}^a$ can be written as

$$\begin{aligned} J_{\mu\text{reg}}^a(x) = -\frac{d}{ds} \left\{ s \operatorname{tr} \left[\gamma_\mu t^a \int d^n x \right. \right. \\ \left. \left. \times K_{-s-1}(x, x; \mathcal{D}(A)) \right] \right\} \Big|_{s=0}, \end{aligned} \quad (12)$$

where $K_s(x, x; \mathcal{D})$ is the meromorphic continuation of the evaluation on $x = y$ of the kernel $K_s(x, y; \mathcal{D})$ associated to the operator $\mathcal{D}^s$. This meromorphic extension is a continuous function, even for $x = y$ if $\operatorname{Re} s < -n$ and it has, at the most, a simple pole at $s = -1$. Note that $K_{-1}(x, y; \mathcal{D}) = G(x, y)$ for $x \neq y$. The diagonal ($x = y$) is precisely the singular support of $G(x, y)$. Whenever m , the order of the operator $\mathcal{D}$, satisfies $m > n$, its Green's function will be continuous everywhere and coincide in the diagonal with $K_{-1}(x, y; \mathcal{D})$. In Eq. (12) the t^a are the generators of the Lie algebra of G and tr denotes both γ matrices' and G matrices' traces.

We shall now state the main result in this work as a proposition which provides the finite part of $K_s(x, x; \mathcal{D})$ as an analytic function of s through the relation

$$\text{Finite part } K_s(x, x; \mathcal{D}) \Big|_{s=-1} = \frac{d}{ds} (K_{s-1}(x, x; \mathcal{D})) \Big|_{s=0}. \quad (13)$$

It is just this expression that appears in Eq. (12).

It remains then to give a method to evaluate (13) and then use relation (12) in order to get the regularized current. It is precisely the following proposition which can be used to this end.

Proposition: Let $\mathcal{D}$ be an elliptic invertible operator of order m on an n -dimensional compact manifold M without boundary, with a ray of minimal growth⁶ and $n > m$. Then the following identity holds:

$$\begin{aligned} \frac{d}{ds} (s K_{s-1}(x, x; \mathcal{D})) \Big|_{s=0} \\ = -\frac{1}{m} \int_{|\xi|=1} \left(\frac{d}{ds} C_{-m+n}(x, \xi; s) \right) \Big|_{s=-1} \frac{d\Omega_\xi}{(2\pi)^n} \end{aligned}$$

$$\begin{aligned} + \left\{ G(x, y) - \sum_{j=0}^{n-m-1} \int \frac{d\xi}{(2\pi)^n} [C_j(x, \xi / |\xi|; -1) \right. \\ \left. \times |\xi|^{-m-j} e^{i\xi(x-y)} - \mathcal{A}_0(x, x-y) \right. \\ \left. + M(x)(\log|x-y| + C(x-y)) \right\} \Big|_{x=y}. \end{aligned} \quad (14)$$

Here

$$C_j(x, \xi; s) = \frac{i}{2\pi} \int_\Gamma \lambda^j b_{-m-j}(x, \xi; \lambda) d\lambda, \quad (15)$$

with Γ being a curve in the complex plane, beginning at ∞ , passing along the ray of minimal growth to a small circle about the origin, then clockwise along the circle and back to ∞ along the ray. In (15), b_j are the Seeley's coefficients⁶ of $\mathcal{D}$, $G(x, y)$ is its Green's function, and $\mathcal{A}_0(x, z)$ is a homogeneous function of degree zero defined as follows:

$$\begin{aligned} \mathcal{A}_0(x, z) = \int \frac{d\xi}{(2\pi)^n} \text{P.V.} [C_{n-m}(x, \xi / |\xi|; -1) \\ - M(x)] |\xi|^{-n} e^{i\xi z}. \end{aligned} \quad (16)$$

Finally,

$$M(x) = \frac{1}{\omega_n} \int_{|\xi|=1} C_{n-m}(x, \xi; -1) d\Omega_\xi, \quad (17)$$

$$\omega_n \left(\log \frac{1}{|z|} - C(z) \right) = \int \frac{d\xi}{(2\pi)^n} \chi(|\xi| \geq 1) e^{-i\xi z}, \quad (18)$$

with χ the characteristic function of $|\xi| \geq 1$ and $C(z)$ regular in a neighborhood of $z = 0$. All Fourier transforms are taken in the sense of distributions and P.V. means principal value.

One should not be discouraged by the apparent complexity of relation (14). One can think of it as giving the finite part of $K_{-1}(x, y; \mathcal{D})$ at the diagonal in terms of a local part depending on the Seeley's coefficients, which can be evaluated very easily for arbitrary $\mathcal{D}$, plus a contribution arising from the regular part of the Green's function (at $x = y$) which, given a particular $\mathcal{D}$, one will be able to or not be able to solve closely. As we shall see below, Eqs. (12)–(14) can be used to obtain currents with great simplicity, and allow for relevant analysis for the case of general field-strength configurations. The term between brackets in (14) is the one where the nonregular part of $G(x, y)$ on the diagonal is subtracted. Concerning the first term, it is generated by the regularization method and it adds to the eventually nonzero regular part of $G(x, x)$ to give the final result. As we shall see in our $n = 3$ example, even without computing the regular part of $G(x, x)$ for arbitrary A_μ , we will manage to extract the relevant physical properties (in particular, parity violation) from (14).

The proof of the Proposition is given as the Appendix. In what follows, examples on how relation (14) can be used are discussed.¹⁰ First consider the very simple case of two-dimensional quantum electrodynamics (discussed by Schwinger in Ref. 2) in order to shed some light on the different contributions to (14).

As we stated above, within the Schwinger approach, one starts from Eq. (6) and the knowledge (in this particular case) of the complete Green's function for an arbitrary background field A_μ .

$$G(x, y) = (-i/2\pi) \exp[\gamma_5(\phi(x) - \phi(y)) + i(\eta(x) - \eta(y))\gamma_\mu[(x_\mu - y_\mu)/|x - y|^2]], \quad (19)$$

where ϕ and η are related to A_μ through the equation

$$A_\mu = -\epsilon_{\mu\nu} \partial_\nu \phi + \partial_\mu \eta. \quad (20)$$

Inserting relations (19) and (20) in Eq. (6) one gets the gauge-invariant expression

$$J_{\mu_{\text{pt}}}(x) = -(1/\pi)[A_\mu - \partial_\mu \Delta^{-1} \partial_\nu A_\nu] \quad (21)$$

(with pt we indicate that the current has been obtained using the proper-time method).

It is interesting to note that the first term in (21) arises from the phase factor while the second one comes from the η exponential in (19). The relation between J_μ and the axial anomaly¹¹ can be understood from (21) if one notes that the (regularized) fermion determinant in $n = 2$ dimensions can be computed as the (anomalous) Jacobian arising from a "decoupling chiral change" of the fermionic variables

$$\psi = e^{\gamma_5 \phi + i\eta} \chi, \quad \bar{\psi} = \bar{\chi} e^{\gamma_5 \phi - i\eta}. \quad (22)$$

One gets¹²

$$\log \frac{\det \mathcal{D}}{\det i\partial} = \log J = -\frac{1}{2\pi} \int A_\mu [\delta_{\mu\nu} - \partial_\mu \Delta^{-1} \partial_\nu] A_\nu d^2x, \quad (23)$$

and (21) can be derived from (23) by simple differentiation. One can easily see that due to the presence of the first term in Eq. (21), the Ward identity for γ_5 transformations is anomalous since in two dimensions,

$$J_\mu^5 = i\epsilon_{\mu\nu} J_\nu \quad (24)$$

and hence

$$\partial_\mu J_\mu^5 = -i\epsilon_{\mu\nu} \partial_\mu A_\nu. \quad (25)$$

Let us now reobtain this result using the method summarized in the Proposition. One has to compute some terms in the asymptotic expansion of the symbol of $\mathcal{D}^{-s}$ with $s = 1$. In particular, for the $n = 2, m = 1$ case we are dealing with, only $(dC_1/ds)(x, \xi, s)$ is needed since all other terms containing C_j can be easily seen to vanish for $s = 0$. Using the identity

$$\int_{|\xi|=1} \frac{dC_1}{ds}(x, \xi, s) \Big|_{s=-1} = \frac{d\Omega_\xi}{(2\pi)^2} = \frac{i}{2\pi} \int_{\Gamma} \frac{d\lambda}{\lambda} \log \lambda \int_{|\xi|=1} b_{-2}(x, \xi, \lambda) \frac{d\Omega_\xi}{(2\pi)^2} \equiv C_1, \quad (26)$$

our problem reduces to the evaluation of the Seeley's coefficient b_{-2} , which can be easily computed (see, for example, Ref. 8)

$$b_{-2}(x, \xi, \lambda) = [1/(\lambda^2 - |\xi|^2)^2] [2\lambda \xi_\mu A_\mu - (\lambda^2 - |\xi|^2)A - 2\xi_\mu A_\mu \xi]. \quad (27)$$

The contribution from C_1 to $J_{\mu_{\text{reg}}}$ can be computed using Eqs. (26), (27), and (14)

$$\text{tr } \gamma_\mu C_1 = + (1/2\pi) A_\mu. \quad (28)$$

From (17) one can show that in this case $M(x) = 0$. Concerning $\mathcal{A}_0(x, z)$ one gets

$$\mathcal{A}_0(x, z) = -(1/4\pi)(A - 2A_\mu z_\mu (z/z^2)), \quad z = x - y, \quad (29)$$

and hence

$$(G(x, y) - \mathcal{A}_0(x, x - y))|_{x=y} = \frac{1}{2\pi} \gamma_\mu \epsilon_{\mu\nu} \partial_\nu \phi + \frac{1}{4\pi} A. \quad (30)$$

Collecting all these results, one finally gets

$$J_{\mu_{\text{reg}}}(x) = -(1/\pi) \epsilon_{\mu\nu} \partial_\nu \phi, \quad (31)$$

which obviously coincides with (21) if one uses the relation $\eta = \Delta^{-1} \partial_\nu A_\nu$. Note that in this last evaluation of J_μ neither a phase factor nor a symmetric limit has to be taken in order to get a finite, gauge-invariant result.

As a second example, let us study the case of fermions coupled to $SU(N)$ gauge fields in $n = 3$ dimensions. This theory has been recently studied in connection with the spontaneous breakdown of a space-time symmetry³⁻⁵: the computation of J_μ for constant or static field strengths using Schwinger's method has shown a parity-violating contribution to the current (here parity means coordinate reflection invariance).

As we shall see, the use of the regularization method we propose shows that parity is violated for arbitrary (that is, not necessarily constant or static) field strengths. Indeed, we show for general A_μ that

$$J_{\mu_{\text{reg}}}^a(x) = \pm (i/16\pi) \epsilon_{\mu\nu\rho} F_{\nu\rho}^a + \dots, \quad (32)$$

where we indicate with dots normal parity terms. We have chosen the Pauli matrices as the 2×2 Dirac matrices and, as usual, we work in Euclidean space. The A_μ takes values in the Lie algebra of $SU(N)$. (Concerning the $\pm$ sign in J_μ , see the discussion below.)

It will be clear from the explicit calculations that the parity-violating term is a product of the (gauge-invariant) regularization method exactly as the preservation of gauge invariance was at the origin of the breakdown of chiral invariances in the $n = 2$ case. That is, preservation of gauge invariance introduces, via regularization, violation of other symmetries (like chiral or parity symmetries). In both cases this violation appears as a topological term. In the present $n = 3$ case, integration of (32) gives a topological mass term (Chern-Simons secondary invariant) for the effective action. Even if absent at the classical level, the necessity of regularization induced this term, first studied by Jackiw and Templeton¹³ (see also Refs. 3, 4, 14, and 15).

Equation (32) is obtained from relations (12)–(14) following the same steps described for the $n = 2$ example. Indeed, one can easily compute the contribution to J_μ from C_2 , defined as

$$C_2 = \int_{|\xi|=1} \frac{d}{ds} C_2(x, \xi, s) \Big|_{s=-1} = \frac{d\Omega_\xi}{(2\pi)^3} = \frac{i}{2\pi} \int_{\Gamma} \frac{d\lambda}{\lambda} \log \lambda \int_{|\xi|=1} b_{-3}(x, \xi, \lambda) \frac{d\Omega_\xi}{(2\pi)^3}, \quad (33)$$

where Seeley's coefficient b_{-3} is given by⁸

$$b_{-3}(x, \xi, \lambda) = \frac{i}{(\lambda^2 - |\xi|^2)^3} \{ 2\xi_\mu \partial_\mu A - 2\lambda \xi_\mu \partial_\mu A - 2\lambda \xi_\mu \partial_\mu A + 2\lambda^2 \xi_\mu \partial_\mu A \}$$

$$\begin{aligned}
& + (\lambda^2 - |\xi|^2) \partial \mathcal{A} \xi - \lambda (\lambda^2 - |\xi|^2) \partial \mathcal{A} \quad (34) \\
& - i \xi \mathcal{A} \xi \mathcal{A} \xi + i \lambda (\xi \mathcal{A} \xi \mathcal{A} + \xi \mathcal{A} \mathcal{A} \xi + \mathcal{A} \xi \mathcal{A} \xi) \\
& - i \lambda^2 (\xi \mathcal{A} \mathcal{A} + \mathcal{A} \xi \mathcal{A} + \mathcal{A} \mathcal{A} \xi) + i \lambda^3 \mathcal{A} \mathcal{A} \}.
\end{aligned}$$

One finally gets

$$C_2 = - (1/8\pi)(\partial \mathcal{A} - i \mathcal{A} \mathcal{A}) + (1/16\pi)(\partial_\mu A_\mu - 2i A_\mu A_\mu). \quad (35)$$

Note that in the computation of C_2 , one does not need the explicit form of A_μ . That is the reason why our result will be valid for arbitrary A_μ . Concerning $\mathcal{A}_0(x, z)$ it is interesting to stress that, as $\mathcal{A}_0(x, iz) = (\text{sgn } i)^n \mathcal{A}_0(x, z)$, no regular contribution to it can arise whenever n is odd. In the present $n = 3$ case it reads

$$\begin{aligned}
\mathcal{A}_0(x, z) &= 2\pi^2 \left[-\frac{1}{2} \partial_\alpha A_\mu (z_\alpha z_\mu / |z|^3) \not{z} + (1/4 |z|) (z_\alpha \gamma_\mu \right. \\
&+ z_\mu \gamma_\alpha + \not{z} \delta_{\alpha\mu}) \partial_\alpha A_\mu - (z_\alpha / |z|) \partial_\alpha \mathcal{A} \left. \right] + A_\nu A_\alpha i \pi^2 \\
&\times [\not{z} (z_\nu z_\alpha / |z|^3) + i \epsilon_{\nu\alpha\beta} (z_\beta / |z|)] + \pi^2 \partial \mathcal{A} (\not{z} / |z|). \quad (36)
\end{aligned}$$

Together with the terms involving C_0 and C_1 in the right-hand side (rhs) of (14), $\mathcal{A}_0(x, z)$ cancels the nonregular part of $G(x, y)$ at the diagonal. Of course, in order to give the complete expression for J_μ one has to compute the contribution of the regular part of $G(x, x)$. This would in principle require the knowledge of $G(x, y)$ for arbitrary $A_\mu(x)$. However, no parity-violating term can arise from this contribution since regular terms, not induced by the regularization method, cannot break an invariance of the theory. Using this fact, it is easy to obtain (32) from Eqs. (33), (12), and (14).

As we stressed above, our result is valid for arbitrary non-Abelian $F_{\mu\nu}$. It confirms the results obtained in Refs. 3 and 4 for the constant and static $F_{\mu\nu}$ cases. One should note that in these last works, there is a sign ambiguity in J_μ as a spur of the regularization procedure. In our approach, this ambiguity can be traced back to the choice of the Γ curve in the λ integrations [see Eq. (15)]. These integrations are of the type

$$I_\Gamma = \int_\Gamma d\lambda \frac{\lambda^q}{(\lambda^2 - 1)^p} \log \lambda, \quad (37)$$

where q depends on the order of the operator and the space-time dimensions and Γ is taken, for example, in the upper half-plane. Had we chosen a curve Γ in the negative half-plane the result should read

$$I_{\bar{\Gamma}} = (-1)^{q+1} I_\Gamma. \quad (38)$$

For the $n = 2$ case, $q = 1$ and there is no ambiguity. On the other hand, for $n = 3$ all integrals correspond to $q = 0$ and hence there is a sign ambiguity due to the arbitrariness of the Γ -curve choice.

Throughout this work we have treated $\mathcal{D}$ as an invertible operator, an assumption which is not in general satisfied. When zero modes are present, one proceeds as follows: one constructs the invertible operator $\mathcal{D} + im$, computes J_μ , and then makes m go to zero. Since Seeley's coefficients are polynomials in m , the (finite) limit coincides with the result obtained with $m = 0$ from the beginning.¹⁶

III. CONCLUSIONS

In summary, we have developed a regularization method based on Seeley's technique for complex powers of pseu-

dodifferential operators. In our approach, the generating functional Z (the fermion determinant) is regularized once and for all using the ζ -function method and this guarantees, as it has been rigorously proven using the ζ -function differentiability properties, that the current obtained from Z by functional differentiation are finite.

In contrast with the point-splitting technique, our prescription is, *per se*, gauge invariant and it does not need the addition of a phase factor *à la* Schwinger. One can also clearly understand the relation between axial anomalies and parity violations^{3,4,15} in our approach: the ζ function provides a definition of the fermionic generating functional (that is, a definition of the path-integral fermionic measure). From this finite generating functional, ground-state currents can be computed; the preservation of gauge invariance then forces violation of chiral symmetry in even dimensions or parity in odd ones. These violations appear as topological terms in current conservation equations (n even) or currents (n odd).

Concerning explicit computations, the evaluation of J_μ (as well as $\partial_\mu J_\mu^5$ for n even⁸) reduces to calculating a few Seeley's coefficients. Although in some cases this can become tedious, it is important to notice that after calculations one can analyze fermion currents in arbitrary background fields.

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APPENDIX: PROOF OF THE PROPOSITION

Since D^s is a pseudodifferential operator of order ms , $K_s(x, y; D)$ can be written, for $\text{Re}(-s)$ sufficiently large,⁶ as $K_s(x, y; D)$

$$= \sum_{j=0}^{n-m} \int \frac{d\xi}{(2\pi)^n} C_j(x, \xi; s) e^{i(x-y)\xi} + K(x, y; R(s)), \quad (A1)$$

where the C_j are the terms of an asymptotic expansion of the symbol of D^s and $K(x, y; R(s))$ is the kernel of an operator $R(s)$, regularizing of order $ms - (n - m + 1)$. Moreover, $R(s)$ is an analytic function of s in a region containing $s = -1$, taking values in the space of continuous linear operators from the Sobolev space $H^{-n/2-\epsilon}(M)$ into $H^{n/2+\epsilon}(M)$, $\epsilon > 0$. From this, we can see that

$$\lim_{y \rightarrow x} \lim_{s \rightarrow -1} K(x, y; R(s)) = \lim_{s \rightarrow -1} \lim_{y \rightarrow x} K(x, y; R(s)). \quad (A2)$$

Note that if $R(s): H^{-n/2-\epsilon}(M) \rightarrow H^{n/2+\epsilon}(M)$, $K(x, y; R(s))$ is continuous even for $x = y$.

Now for $x \neq y$, $G(x, y) = \lim_{s \rightarrow -1} K_s(x, y; D)$. Then

$$K(x, y; R(-1)) = G(x, y) - \sum_{j=0}^{n-m} \int \frac{d\xi}{(2\pi)^n} C_j(x, \xi; -1) e^{i\xi(x-y)}, \quad (A3)$$

even for $x = y$. Now, from (A3) we have the following identity:

$$\begin{aligned} \lim_{y \rightarrow x} \left[G(x, y) - \sum_{j=0}^{n-m} \int \frac{d\xi}{(2\pi)^n} C_j(x, \xi; -1) e^{i\xi(x-y)} \right] \\ = \lim_{y \rightarrow x} K(x, y; R(-1)) = \lim_{s \rightarrow -1} K(x, x; R(s)) \\ = \lim_{s \rightarrow -1} \left[K_s(x, x; D) - \sum_{j=0}^{n-m} \int \frac{d\xi}{(2\pi)^n} C_j(x, \xi; s) \right]. \quad (\text{A4}) \end{aligned}$$

If we now split the ξ integration and use the fact that $C_j(x, \xi; s)$ is a homogeneous function of degree $ms - j$ in ξ for $|\xi| > 1$ to write the identity

$$\begin{aligned} \int_{|\xi| > 1} C_j(x, \xi; s) \frac{d\xi}{(2\pi)^n} \\ = \frac{-1}{ms - j + n} \int_{|\xi| = 1} C_j(x, \xi; s) \frac{d\xi}{(2\pi)^n}, \quad (\text{A5}) \end{aligned}$$

we have

$$\begin{aligned} \lim_{y \rightarrow x} \left[G(x, y) - \sum_{j=0}^{n-m} \int \frac{d\xi}{(2\pi)^n} C_j(x, \xi; -1) e^{i\xi(x-y)} \right] \\ = \lim_{s \rightarrow -1} \left\{ K_s(x, x; D) - \left(\frac{-1/m}{(s+1)} \int_{|\xi|=1} C_{n-m}(x, \xi; s) \frac{d\xi}{(2\pi)^3} \right. \right. \\ \left. \left. + \int_{|\xi| < 1} C_{n-m}(x, \xi; s) d\xi \right) - \sum_{j=0}^{n-m-1} \frac{1}{(2\pi)^n} \left(\frac{-1}{ms + n - j} \right. \right. \\ \left. \left. \times \int_{|\xi|=1} C_j(x, \xi; s) d\xi + \int_{|\xi| < 1} C_j(x, \xi; s) d\xi \right) \right\}. \quad (\text{A6}) \end{aligned}$$

Then, calling

$$\tilde{C}_j(x, \xi) = |\xi|^{-m-j} C_j(x, \xi/|\xi|; -1),$$

Eq. (A6) can be rewritten as

$$\begin{aligned} \lim_{y \rightarrow x} \left[G(x, y) - \sum_{j=0}^{n-m-1} \int \frac{d\xi}{(2\pi)^n} \tilde{C}_j(x, \xi) e^{i\xi(x-y)} \right. \\ \left. - \int \frac{d\xi}{(2\pi)^n} C_{n-m}(x, \xi; -1) e^{i\xi(x-y)} \right] \\ = \lim_{s \rightarrow -1} \left[K_s(x, x; D) - \frac{-1/m}{s+1} \int_{|\xi|=1} C_{n-m}(x, \xi; s) \frac{d\xi}{(2\pi)^n} \right. \\ \left. - \int_{|\xi| < 1} C_{n-m}(x, \xi; s) \frac{d\xi}{(2\pi)^n} \right]. \quad (\text{A7}) \end{aligned}$$

Taking the expression of $C_{n-m}(x, \xi; s)$ in powers of $(s+1)$ we can write (A7) in the form

$$\begin{aligned} \lim_{s \rightarrow -1} \left[K_s(x, x; D) - \frac{1}{s+1} \left(-\frac{1}{m} \right) \int_{|\xi|=1} C_{n-m}(x, \xi; -1) \frac{d\xi}{(2\pi)^n} \right] \\ = -\frac{1}{m} \int_{|\xi|=1} \frac{d\xi}{(2\pi)^n} \frac{d}{ds} C_{n-m}(x, \xi; s) \Big|_{s=-1} \\ + \int_{|\xi| < 1} C_{n-m}(x, \xi; -1) \frac{d\xi}{(2\pi)^n} + \lim_{y \rightarrow x} \left[G(x, y) \right. \\ \left. - \sum_{j=0}^{n-m-1} \int \frac{d\xi}{(2\pi)^n} C_j(x, \xi; -1) e^{i\xi(x-y)} - \int \frac{d\xi}{(2\pi)^n} \right. \\ \left. \times C_{n-m}(x, \xi; -1) e^{i\xi(x-y)} \right]. \quad (\text{A8}) \end{aligned}$$

Then, using¹⁷

$$\begin{aligned} \int_{|\xi| > 1} e^{i\xi z} C_{n-m}(x, \xi; -1) \frac{d\xi}{(2\pi)^n} \\ = \text{P.V.} \int \frac{d\xi}{(2\pi)^n} e^{i\xi z} |\xi|^{-n} \left[C_{n-m}\left(x, \frac{\xi}{|\xi|}; -1\right) - M(x)\chi \right] \\ + M(x) [\log |z| + C(z) + g(z)], \end{aligned}$$

with $g(z)$ and $C(z)$ regular functions and $g(0) = 0$, we get Eq. (14).

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